



PROGRESS MONITORING FORM

Activity Title: User Interface and Database Integration Laboratory

Course/Subject: Integrative Programming

Instructor: LITODANE L. LOÑEZ

Student Name/s: MACATANTAN, PRINCESS JHOANA B.

MANANGAN, ZYRON T.

SARMIENTO, ARCHIE P.

DOMDOM, JERWAYNE A.

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Section/Year: BSIT – WMT III-2

A. OBJECTIVES

- ☐ Create a database using MySQL
- ☐ Establish connection between GUI and database
- ☐ Perform CRUD operations (Create, Read, Update, Delete)
 - All forms shall be included such as Add New User, Edit User or any other forms that will be present in your system. All forms should be labeled according to their functions. Minimum: 7 forms ➤ List down the names of the forms here:

Form Name/Label	Function
SignInForm	Authenticates users and grants access to the system by verifying their credentials against the database.
SignUpForm	Registers new user accounts and securely saves their information and hashed passwords into the database.
Dashboard	Serves as the main landing page after a successful login, providing an overview and navigation to other modules.



AddStudentForm	Collects and inserts new student records (personal, academic, and guardian information) into the database.
AddCourseForm	Collects and inserts new course details into the database.
CourseForm	Acts as a management page for courses and contains navigation buttons to switch between different system modules.
StudentsForm	Displays and manages the records of enrolled students
EditStudentForm	This form is used to modify existing student records. It is triggered when the Edit button is clicked on the StudentsForm. It automatically populates fields with the selected student's data for easy updating.
EditCourseForm	This form allows administrators to modify existing course details. It is accessed by selecting a course in the CoursesForm and clicking the Edit button. It enables updates to the Course Code, Name, Instructor, and Schedule.
lblStudentsCounter	Located on the Dashboard, this label displays the total count of students currently enrolled in the database. It uses the SELECT COUNT(*) SQL command to provide real-time statistics.
lblCoursesCounter	This label on the Dashboard displays the total number of courses offered. Like the student counter, it updates dynamically by querying the Courses table.

- ☐ Design a functional GUI interface
- ☐ Apply proper coding practices such as validations
 - Add further validations in the CRUD operators most specifically on the part of Delete and Update.



➤ Dialog boxes is highly recommended.

☐ Deadline of Submission will be on **May 04, 2026**. Face to Face presentation

B. TASK PROGRESS CHECKLIST

(Kindly check your accomplishments for each day. Check only the tasks you have completed, then upload this to MS Teams.)

Task Description	Status (✓/X)	Remarks
Database created in phpMyAdmin	✓	Created the local db_sis database using Laragon/MySQL.
Table structure defined correctly	✓	Defined tables for Students, Courses, Departments, YearLevels, and Users with appropriate primary keys and data types.
Connection established (MySQL Connection)	✓	Successfully configured MySqlConnection to connect the C# WinForms application to the local server.
Insert (Create) function implemented	✓	Implemented parameterized INSERT queries for the Add Student, Add Course, and Sign Up forms (including SHA256 password hashing).
Retrieve (Read) data displayed in GUI	✓	Successfully implemented real-time data fetching from the MySQL database to the DataGridView.
Update function working	✓	The system accurately modifies existing records and reflects changes in the database immediately.
Delete function working	✓	Data removal is fully operational with a confirmation prompt to ensure data integrity.
GUI design completed	✓	The user interface is finalized, featuring a responsive sidebar navigation and a professional color scheme.



Error handling implemented	✓	Added field validation, password matching checks, duplicate entry prevention, and try-catch blocks for <code>MySqlConnection</code> .
Program runs without errors	✓	Resolved event wiring issues and cleared build errors for smooth execution.

C. STUDENT REFLECTION

(What did you learn? Challenges encountered? Per Student)

Student 1: Zyron T. Manangan

Setting up the database in phpMyAdmin taught me that a well-organized table structure is the backbone of any successful Information Management System. My primary focus was ensuring that all data types were correctly defined to prevent structural errors, which eventually allowed for seamless real-time data fetching from MySQL to the C# DataGridView. The most rewarding aspect of the project was implementing the dynamic dashboard counters, such as `lblStudentsCounter`, which successfully display accurate enrollment statistics directly from the backend.

Student 2: Princess Jhoana B. Macatantan

I realized throughout this development process that a solid back-end foundation is essential for building a functional and reliable system. Planning the table structure required careful attention to detail to ensure data integrity, especially when handling sensitive operations like "Update" and "Delete". Seeing the "Student updated successfully!" notification was proof that our meticulous planning during the design phase effectively translated into a stable and responsive user experience.

Student 3: Archie P. Sarmiento

Having moved past the initial conceptual stages, I successfully tackled the database connection and CRUD operations, which I originally identified as my biggest technical challenges. I gained a deep understanding of how to use the `MySqlConnection` and `DataTable` to bridge the gap between our database and the graphical user interface. Furthermore, completing the delete function with an integrated confirmation safety check taught me the importance of prioritizing user security and data validation in software development.

Student 4: Jerwayne A. Domdom

I have transitioned from simply understanding system requirements to actively writing the code that facilitates communication between the application and the database. While learning to save, modify, and remove information was the most difficult part of the journey, I successfully mastered the use of SQL commands like `INSERT`, `UPDATE`, and `DELETE`. Finalizing the GUI design, including the sidebar navigation and professional dashboard, helped me realize how critical a clean front-end is for providing a better overall user experience.



D. SCREENSHOTS

(Check the tasks you completed each day. Take screenshots as proof of your work.)

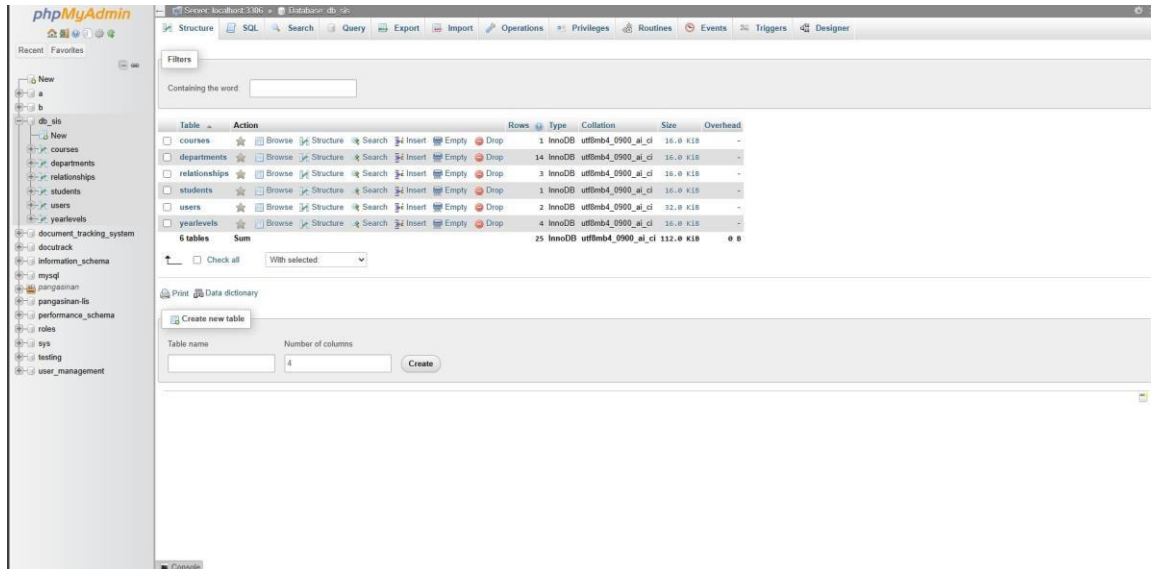


Figure 1: Creation of the local database db_sis for the Student Information System.

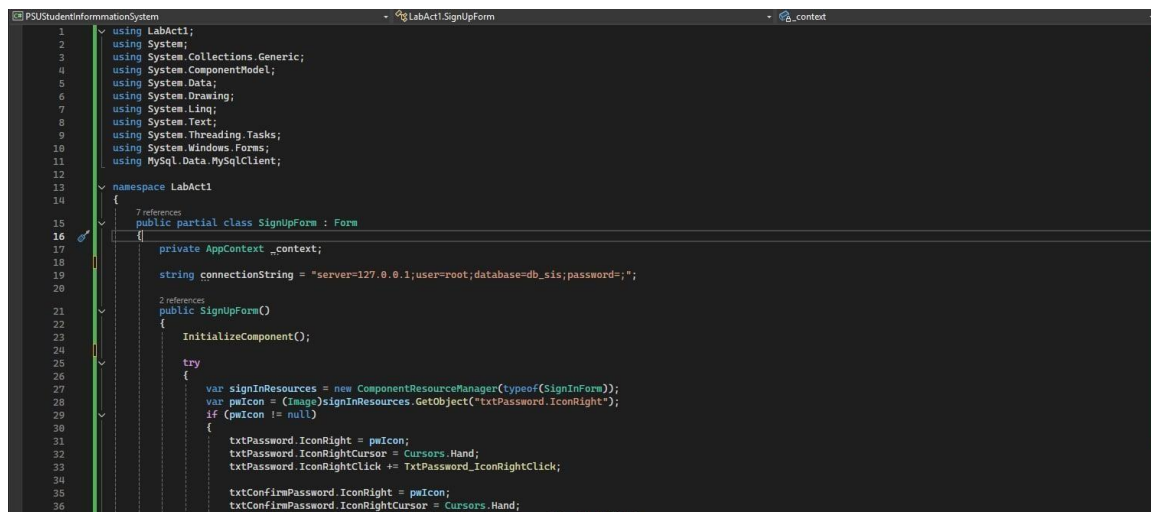


Figure 2: Successful establishment of the MySQL database connection within the C# application.



#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 StudentID	int			No	None		AUTO_INCREMENT	Change Drop More
☐	2 FirstName	varchar(50)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	3 LastName	varchar(50)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	4 MiddleName	varchar(50)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
☐	5 Suffix	varchar(10)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
☐	6 BirthDate	date			No	None			Change Drop More
☐	7 Gender	varchar(10)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	8 Email	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	9 PhoneNumber	varchar(20)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	10 Address	varchar(255)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	11 YearLevel	varchar(20)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	12 Department	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	13 GuardianName	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	14 GuardianRelationship	varchar(50)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	15 GuardianEmail	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
☐	16 GuardianPhone	varchar(20)	utf8mb4_0900_ai_ci		No	None			Change Drop More

Figure 3: Database table structure and naming conventions for the Students table.

Pangasinan State University
Student Information System

Create an Account

First Name *

Zyron

Last Name *

Manangan

Middle Name (optional)

Suffix (optional)

Jr, Sr, III

Success

Registration Successful!

OK

Password *

zeiii123

Confirm Password *

zeiii123

Cancel

Create Account

Figure 5: Successful execution of the Insert (Create) function during a new user registration.

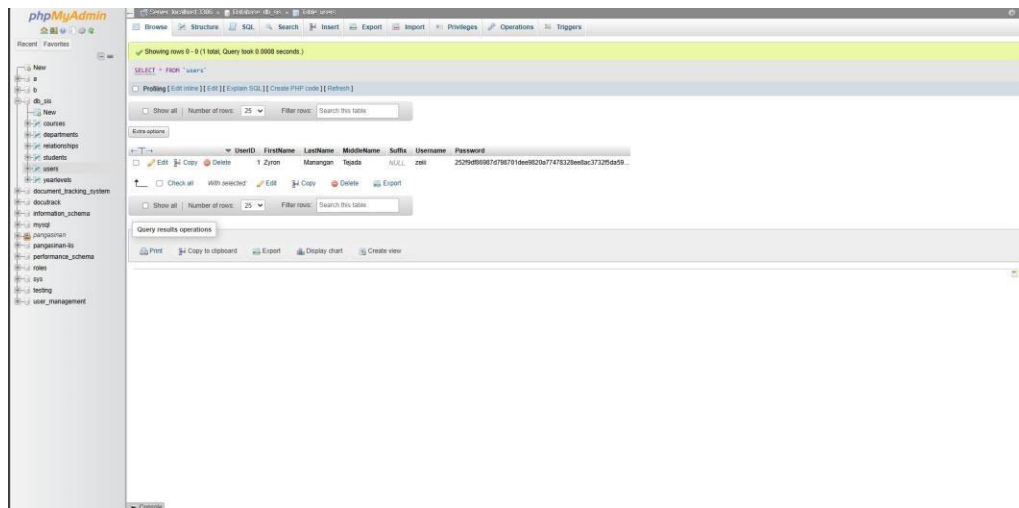


Figure 6: Proof of the implemented Create functionality, showing the newly registered user successfully inserted into the users table.

The screenshot shows the 'Student Registration' form. The form is divided into several sections: 'Personal Information', 'Academic Information', and 'Guardian Information'. The 'Personal Information' section includes fields for First Name, Last Name, Middle Name (optional), Suffix (optional), Birth Date, Gender, Email, and Phone Number. The 'Academic Information' section includes fields for Year Level and Department. The 'Guardian Information' section includes fields for Guardian Name, Relationship, Email, and Phone Number. A 'Success' dialog box is displayed in the center, stating 'Student successfully added!' with an 'OK' button. The form has 'Cancel' and 'Add Student' buttons at the bottom right.

Figure 7: System confirmation displaying the successful insertion of a new student record, including personal, academic, and guardian details.



The screenshot shows the phpMyAdmin interface with the 'students' table selected. The table contains two records:

StudentID	FirstName	LastName	MiddleName	Suffix	BirthDate	Gender	Email	PhoneNumber	Address	YearLevel	Department	GuardianName	GuardianRelationship	GuardianEmail	GuardianPhone
1	Princess	Macatantan	NULL	NULL	2005-05-23	Female	princessmacatantan@gmail.com	99962967995	Real, Unibonda, Pangasinan	3rd Year	BS Information Technology	Joan Macatantan	Parent	NULL	99394596796
2	Zyon	Manangan	Tajada	NULL	2005-09-29	Male	zyonmanangan@gmail.com	99394596795	Nalisan Norte, Manangul, Pangasinan	3rd Year	BS Information Technology	Melinda Manangan	Parent	NULL	99705783335

Figure 8: Database view confirming the successful insertion of the new student record into the students table.

The screenshot shows the 'Add Course' form with the following fields filled:

- Course Name: Captone Project 1
- Course Code: (empty)
- Department: BS Information Technology
- Schedule: Mon-Wed 2:00-3:00PM

A success message dialog box is displayed, stating: "Success Course successfully added!". The dialog has an "OK" button.

At the bottom of the form, there are "Cancel" and "Add Course" buttons.

Figure 9: Successful execution of the Create function within the Add Course form.

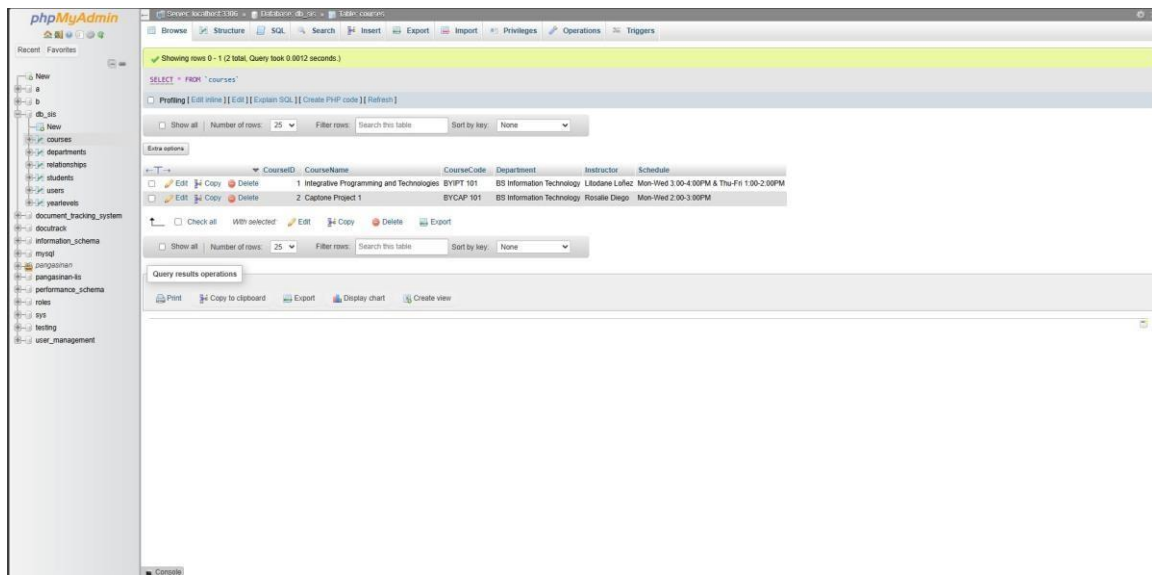


Figure 10: Database view confirming the successful insertion of the new course record into the courses table.

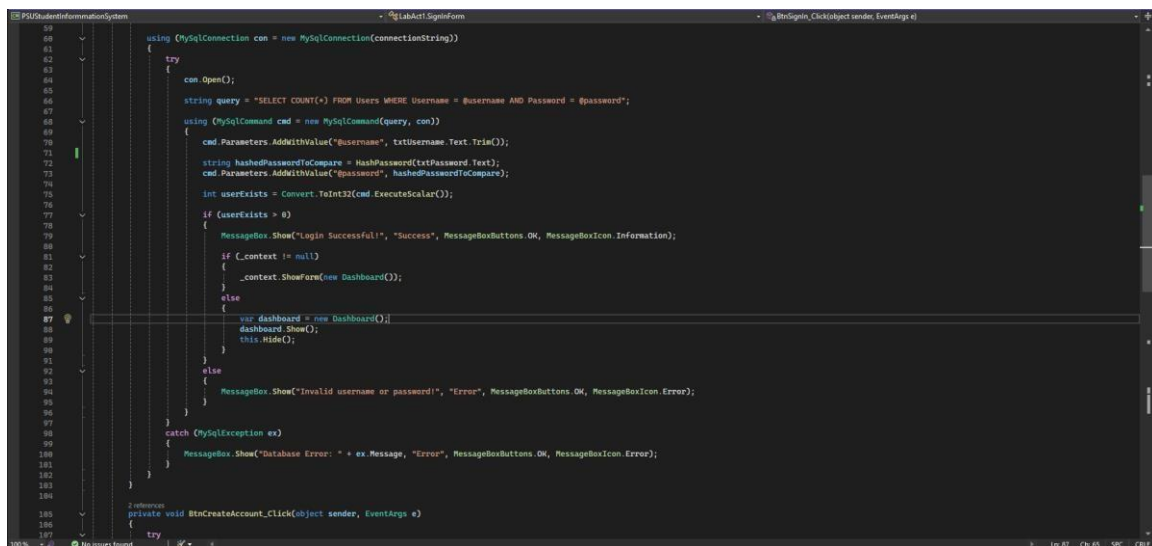


Figure 11: Implementation of a simple try-catch block for database error handling.

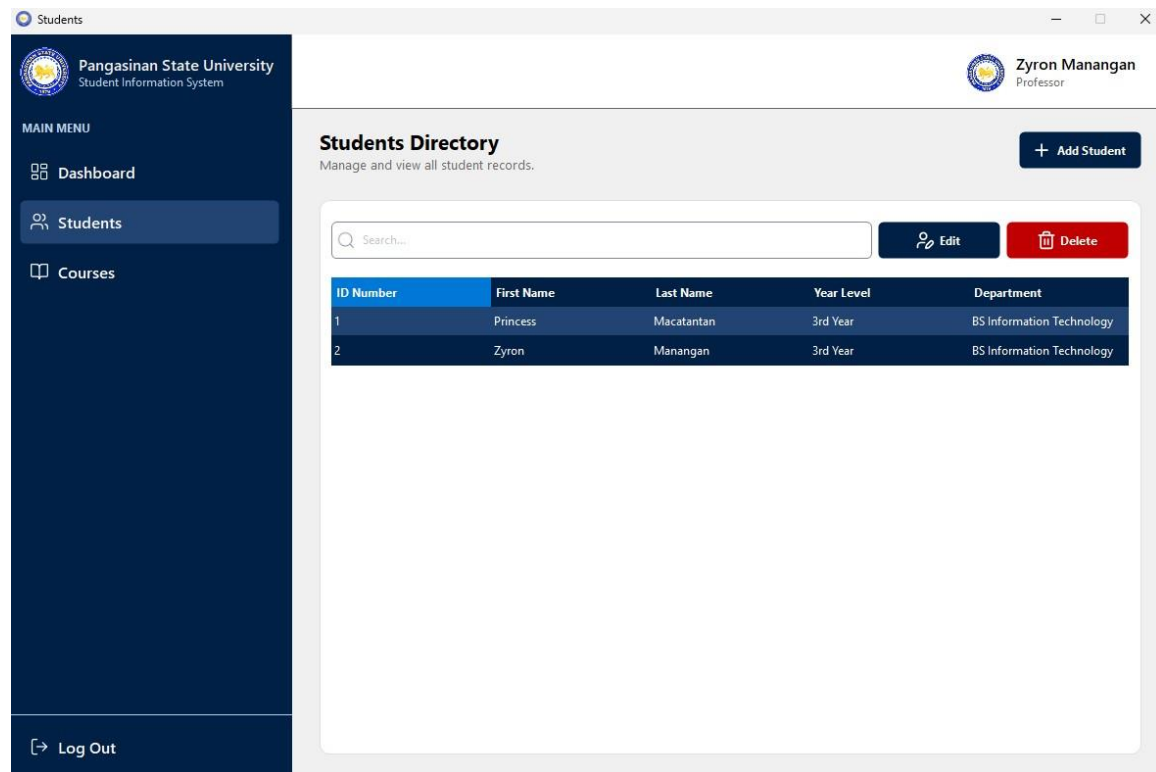


Figure 12. Front-end view of the student management module showing the data retrieval functionality.

```
using System;
using System.Data;
using System.Data.SqlClient;
using System.Windows.Forms;

namespace PangasinanStateUniversity
{
    public partial class Form1 : Form
    {
        private void LoadStudentsData()
        {
            string connectionString = "server=127.0.0.1;user=root;database=db_sis;password=";
            using (SqlConnection con = new SqlConnection(connectionString))
            {
                try
                {
                    con.Open();
                    string query = "SELECT StudentID AS 'ID Number', FirstName AS 'First Name', LastName AS 'Last Name', YearLevel AS 'Year Level', Department FROM students";
                    using (SqlCommand cmd = new SqlCommand(query, con))
                    using (MySqlDataAdapter adapter = new MySqlDataAdapter(cmd))
                    {
                        dtStudents = new DataTable();
                        adapter.Fill(dtStudents);
                        dgvStudents.DataSource = dtStudents;
                    }
                }
                catch (Exception ex) { MessageBox.Show("Error: " + ex.Message); }
            }
        }
    }
}
```

Figure 13. Backend logic for populating the dgvStudents DataGridView with real-time database information.





Pangasinan State University
Student Information System

Student Registration

Personal Information

First Name *	Last Name *
Zyron HAHHA	Manangan
Middle Name (optional)	Suffix (optional)
Tejada	Dela Cruz
Birth Date *	Gender *
Tuesday, September 20, 2005	☒ Male ☐ Female
Email *	Phone Number *
zyronmanangan@	09394606766
Address *	
Nalsian Norte, Ma	

Academic In

Year Level *	Department *
3rd Year	BS Information Technology

Guardian Information

Guardian Name *	Relationship *
Melinda Manangan	Parent
Email	Phone Number *
user@example.com	09705783336

Success

 Student updated successfully!

Figure 14. Demonstration of the system's ability to update and save changes to existing student profiles.



The screenshot shows the 'Courses' management interface of the Pangasinan State University Student Information System. The interface includes a sidebar with a 'MAIN MENU' containing 'Dashboard', 'Students', and 'Courses' (the active page). The top right shows the user as 'Unknown User Professor'. The 'Courses' section has a subtitle 'Manage classes, instructors, and enrollments.' and an '+ Add Course' button. Below this is a search bar and 'Edit' and 'Delete' buttons. A table lists three courses:

ID	Course Code	Course Name	Department	Instructor	Schedule
1	BYIPT 101	Integrative Programmi...	BS Information Techn...	Litodane Loñez	Mon-Wed 3:00-4:00P...
2	BYCAP 102	Capstone Project 2	BS Information Techn...	Rosalie Diego	Mon-Wed 2:00-3:00PM
3	BYTE...	Confirm	BS Information Techn...	Desire Licawen	Mon-Wed 1:00-2:00PM

A 'Confirm' dialog box is overlaid on the table, asking 'Delete this course?' with 'Yes' and 'No' buttons.

Figure 15. Implementation of the "Confirm Deletion" safety dialog in the user interface.



E. SIGNATURES

Student Signature/s:

ZYRON T. MANANGAN

PRINCESS JHOANA B. MACATANTAN

ARCHIE P. SARMIENTO

JERWAYNE A. DOMDOM

Instructor Signature: _____